# MILITARY SPECIFICATION
# ACTUATOR, CREW EJECTION 1000-36
# ASSEMBLY

## 1. SCOPE

1.1 This specification is the applicable detail specification for Actuator, Crew Ejection 1000-36 Assembly, loaded with cartridge 5000-1, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

| | |
|---|---|
| MIL-A-2550 | - Ammunition and Special Weapons, General Specification For |

STANDARDS
Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-1167 | - Ammunition Data Card |
| MS-28741 | - Hose Assembly - Detachable End Fitting, Medium Pressure |

DRAWINGS
Munitions Command

| | |
|---|---|
| C9218443 | - Carton, Packing, Ammunition, for Rotary Actuator, 1000 or 1925 Series |
| C9218444 | - Box, Packing Ammunition, for Rotary Actuator, 1000 or 1925 Series |

Talley Industries

| | |
|---|---|
| 1000-36 | - Rotary Actuator Assembly |

FSC 1377

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(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Breakaway torque.- All threaded joints of the assembled Actuator shall withstand a breakaway torque application of 200 inch pounds without loosening or deformation.

3.3 Residual magnetism.- The Actuator shall not deflect the indicator of the approved compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Initial lock security.- The lock pin shall secure the piston component of the ball screw assembly so that the strap drum component is not free to rotate.

3.5 Functional.

3.5.1 The actuator shall not fail to fire.

3.5.2 The actuator shall function without mechanical failure (See 6.1).

3.5.3 Initiation.- The Actuator shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, Size 4, Standard MS28741, at $70^{\circ} \pm 5^{\circ}$F.

3.5.4 Ballistics.- The Actuator loaded with an accepted cartridge and temperature conditioned, when fired, shall separate and eject a 400 pound load from the approved aircraft seat within an operating time of .150 seconds minimum and .500 seconds maximum, and function within the performance limits specified in Table I.

TABLE I

| Temperature | Pressure (psi) | Acceleration (g) |
|---|---|---|
| $65^{\circ}$ to $75^{\circ}$F | 2000 to 4800 | 0.85 to 3.0 |

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defects.



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3.6.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All nonconforming material shall be rejected. Four power magnification may be used in performing visual examination for inspection of such defects as burr and foreign matter.



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4.2.2 (Cont'd)

TABLE II

| Lot Size | Sample Size | Major Accept | Major Reject | Minor Accept | Minor Reject |
|---|---|---|---|---|---|
| 1 - 300 | 100% Examination | - | - | - | - |
| 301 - 800 | 50 | 0 | 1 | 1 | 2 |
| 801 - 1,800 | 90 | 1 | 2 | 2 | 3 |
| 1801 - 10,000 | 200 | 2 | 3 | 3 | 4 |

4.2.2.1 Cartridge Assembly, Drawing 5000-1, covering a detail of Drawing 1000-36.

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| 1. Mixed cartridges | Visual |
| 2. Metal defective | Visual |
| 3. Closure cap missing, damaged or improperly sealed | Visual |
| 4. Marking missing, incorrect or illegible | Visual |
| MAJOR | |
| 101. Foreign matter | Visual |
| MINOR | |
| 201. Case scratched, wrinkled, bulged or dented | Visual |
| 202. Protective finish, missing or inadequate | Visual |

4.2.2.2 Assembly, Drawing 1000-36 with End Closure Assembly (Drawing #1000-132) removed.

| | |
|---|---|
| CRITICAL | |
| 1. Lock pin loose | Manual/Visual |
| 2. Ball screw and piston assembly loose | Manual |
| 3. Blowout plug loose | Manual/Visual |
| MAJOR | |
| 101. Thread damaged | Visual |
| 102. Foreign matter | Visual |
| MINOR | |
| None defined | |



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4.2.2.3 Assembly Drawing 1000-36 with Breech Assembly (Drawing #1022) and back up washer (Drawing #1017) removed.

| Defect | Method of Inspection |
|---|---|
| **CRITICAL** | |
| 1. Cartridge loose | Manual/Visual |
| 2. Cartridge Retainer loose | Manual/Visual |
| 3. Foreign matter in gas port | Visual |
| **MAJOR** | |
| 101. Thread damaged | Visual |
| **MINOR** | |
| None defined | |

4.2.2.4 Assembly Drawing 1000-36

| | |
|---|---|
| **CRITICAL** | |
| 1. Any part missing except as otherwise classified | Visual |
| 2. Marking missing, incorrect or illegible | Visual |
| **MAJOR** | |
| 101. Length overall | Gage |
| 102. Length of strap extended | Gage |
| 103. Length of strap retracted | Gage |
| 104. Shear studs loose | Visual and Manual |
| 105. Cap screw loose | Visual and Manual |
| 106. Threads damaged | Visual |
| **MINOR** | |
| 201. Location labels incorrect | Visual |
| 202. Shipping plug missing | Visual |
| 203. Protective finish inadequate or defective | Visual |

4.2.2.5 Packing in Fiber Box, Drawing C9218443

| | |
|---|---|
| **CRITICAL** | |
| None defined | |



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4.2.2.5 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| **MAJOR** | |
| None defined | |
| **MINOR** | |
| 201. Sealing strips missing, loose or improperly applied | Visual |
| 202. Contents loose | Manual |
| 203. Flaps loose or distorted | Visual |
| 204. Cut or damaged through all impregnated layers | Visual |
| 205. Marking, missing incorrect or illegible | Visual |

4.2.2.6 Packing in Wooden Box, Drawing C9218444

| | |
|---|---|
| **CRITICAL** | |
| None defined | |
| **MAJOR** | |
| None defined | |
| **MINOR** | |
| 201. Contents loose | Manual |
| 202. Hardware missing, loose, broken, or improperly engaged | Manual/Visual |
| 203. Strapping missing, loose, broken or improperly assembled | Manual/Visual |
| 204. Contents of box exposed | Visual |
| 205. Marking missing, incorrect or illegible | Visual |
| 206. Number of units incorrect | Visual |
| 207. Car seal missing, unsealed or improperly positioned | Visual |

4.2.3 Inspection.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- Each actuator shall be subjected to the breakaway torque application. Failure of any actuator to comply with the requirement of 3.2 shall be cause for rejection of the assembly.



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4.2.3.2 Residual magnetism.- Each actuator shall be subjected to the test for excessive residual magnetism. Failure of any actuator to comply with the requirements of 3.3 shall be cause for rejection of the actuator.

4.2.3.3 Initial lock security.- Each actuator shall be subjected to an initial lock test. Failure of any actuator to comply with the requirements of 3.4 shall be cause for rejection of the assembly.

4.2.3.4 Functional tests.- The following tests shall be performed by the Government:

4.2.3.4.1 Ballistics.- Twenty-two units shall be tested as specified in 4.3.4 to determine compliance with the requirements of 3.5.1 to 3.5.3. The units shall be temperature conditioned at 65° to 75°F as provided in 4.3.4.1.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment approved by the Government.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench, set at the torque requirements specified in 3.2 shall be used in conjunction with an adapter to determine the adequacy of the joints. The actuator shall be secured in a holding device and the breakaway torque applied to the end closure and breech assemblies.

4.3.2 Initial lock security.- The actuator shall be tested for initial lock security by securing the actuator body in a holding device and manually pulling on the strap assembly to assure that the webbing cannot be extended or retracted.

4.3.3 Residual magnetism.- The actuator shall be tested for excessive residual magnetism, in an area free of local magnetic effects, by placing the actuator six inches from and at the same height as the compass in a north-south horizontal position (breech end at south) with the compass set in an east-west heading as shown in Figure I. The actuator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly past the compass. This procedure shall be repeated for each 90 degree rotation of the actuator.

4.3.4 Functional tests.

4.3.4.1 Temperature conditioning.- Prior to conducting the ballistic test specified in 4.3.4.3, the actuator samples shall be subjected to the temperature condition specified in 3.5.4. A simulated actuator loaded with a dummy cartridge containing a temperature sensitive element shall be placed in the temperature conditioning chamber at the same time as the actuator samples. The conditioning time shall be a minimum of one hour longer than



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the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.3.4.1.1 Upon completion of the required temperature conditioning, each actuator sample shall be ballistically tested within 5 minutes after removal from the conditioning chamber. Samples shall be reconditioned if this time is exceeded.

4.3.4.2 Method of testing.- The actuator tested under the conditions specified herein shall be mounted on an aircraft seat in the proper position. Install a fixture on the aircraft seat. The fixture shall be designed to cover at least 80% of the area comprising the bottom and back of the aircraft seat. Removable weights may be incorporated in the fixture so that the weight of the ejected mass can be adjusted from 100 to 400 pounds. Mount an accelerometer on the fixture to record acceleration of the ejected mass on a line which bisects the angle between the seat bottom and the seat back. A recording system shall be used to determine the amount of time required to eject the mass after actuation of the actuator under load conditions.

4.3.4.3 Ballistic test.- Utilize the test fixture specified in 4.3.4.2. The weights shall be installed in the fixture and secured to prevent shifting. An M3A1 initiator or simulated method shall be used to actuate the actuator under test. Pressure sensitive devices shall be used to measure the pressure exerted to actuate the actuator and to measure the amount of ballistic gas pressure in the actuator during operation. A pressure port shall be installed in the actuator end closure assembly for recording the ballistic gas pressure. The pressure sensitive device for measuring the amount of pressure used to actuate the actuator shall be inserted into the system by means of a Tee coupling between the actuator and the actuating device. The actuator sample shall be tested in accordance with the functional requirements of 3.5 within the time specified.

## 5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.- Level A.- Packing and marking shall be in accordance with drawings C9218443 and C9218444.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

## 6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

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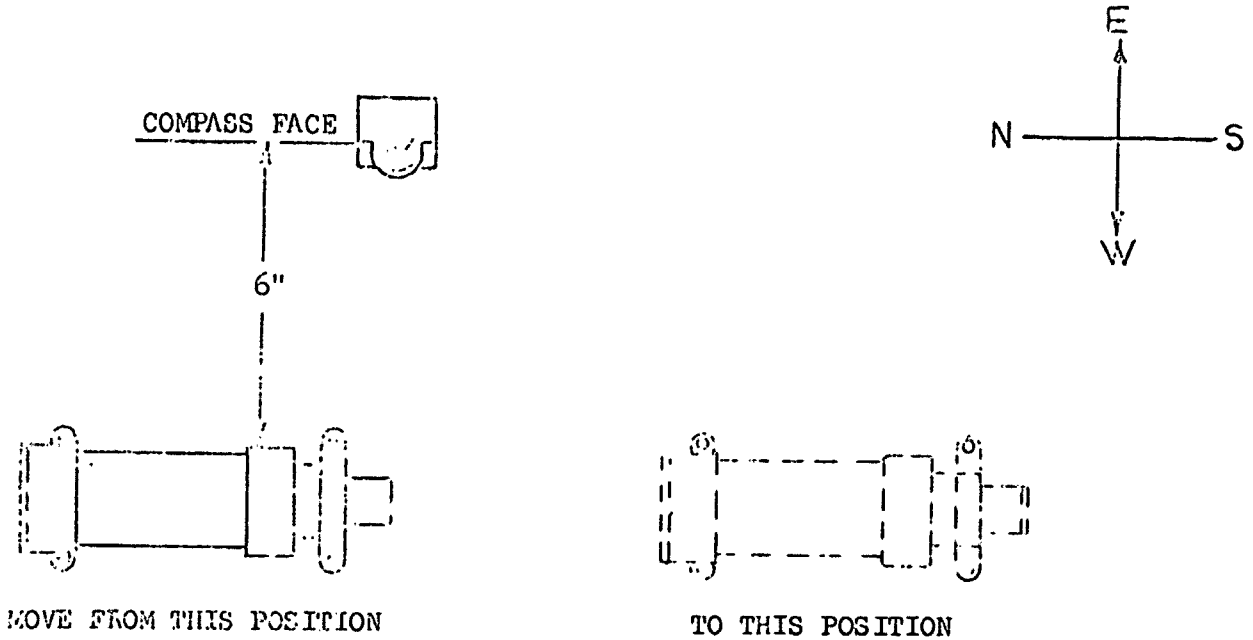
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PILOT'S STANDBY COMPASS, TYPE AN-5766




PLAN VIEW

NOTE: ACTUATOR, ROTARY, ASSEMBLY AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN ACTUATOR, ROTARY, ASSEMBLY 1000-36

FIGURE 1



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# SPECIFICATION ANALYSIS SHEET

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**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

| ORGANIZATION | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
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